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Forecasting International Trade Activity using Oil Prices and Macroeconomic Indicators: A Machine Learning Analysis of G20 Economies

Technical Report

National College of Ireland

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Signature:	John Selvin Raja John
Date:	02/Aug/2026

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Glossary, Acronyms, Abbreviations and Definitions

Term / Acronym	**Definition**
Ablation Analysis	An experiment in which one feature is removed and the model is retrained to assess the feature's contribution
API	Application Programming Interface
Bn	Billions
Chronological Holdout	A later time period reserved for testing so that the model is evaluated on observations occurring after the training data
CPI	Consumer Price Index
Dense Layer	A neural-network layer in which each unit is connected to every unit in the previous layer
Dropout	A regularisation method that temporarily removes a proportion of neural-network units during training to reduce overfitting
Early Stopping	A training method that stops model fitting when validation performance no longer improves
Empirical coverage	The percentage of actual observations that fall within the estimated uncertainty ranges
Estimator	An individual model used within an ensemble, such as one decision tree within XGBoost or Random Forest
Epoch	One complete pass through the full training dataset during the training of a machine-learning or neural-network model
expm1	The transformation $e^x - 1$, used to return predictions from the logarithmic scale to their original values.
FRED	Federal Reserve Economic Data

Term / Acronym	Definition
G20	Group of Twenty
Hyperparameter	A model setting selected before training, such as learning rate, maximum depth or number of estimators
Input Sequence	An ordered group of observations provided to a sequence model, such as 12 consecutive months of predictor values
Lag feature	A predictor created from the value of a variable during an earlier time period
Learning Rate	A setting that controls the size of the adjustments made to a model during training
log1p	The transformation log(1 + x) , used to reduce skewness while allowing zero values
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
One-hot encoding	A method that converts categorical variables into separate binary columns
Pooled model	A single model trained using observations from multiple countries or trade flows.
R^2	Coefficient of determination
Random seed	A fixed starting value used to support reproducible model results
ReLU	Rectified Linear Unit
RMSE	Root Mean Squared Error
Scenario analysis	An experiment that changes selected input values while keeping the remaining predictors unchanged
SHAP	SHapley Additive exPlanations
SVR	Support Vector Regression
Time Step	One position within an input sequence, representing one monthly observation in the LSTM model
Trainable Parameters	The Internal model values adjusted during training to reduce prediction error
Tree-percentile uncertainty range	A lower and upper range calculated from percentiles of predictions generated by individual Random Forest trees
UN Comtrade	United Nations Comtrade database
Validation dataset	Data used to select model settings or determine the appropriate number of training epochs
Validation Loss	The prediction error calculated using validation data during model training.
XGBoost	Extreme Gradient Boosting

Executive Summary

International trade is influenced by seasonal patterns, commodity prices and wider macroeconomic conditions, making accurate forecasting a challenging task. This project developed a machine learning pipeline to forecast monthly import and export values for 18 G20 economies using historical trade data together with oil prices, commodity indices and selected macroeconomic indicators.

Monthly trade data were obtained from UN Comtrade [1] and combined with external economic variables from publicly available sources. The dataset was split chronologically, with data before 2024 used for training and 2024 reserved as the test period. Several machine learning models were evaluated, including Linear, ensemble, boosting and deep learning approaches.

Among all models, the Random Forest with seasonal and external predictors produced the best overall results, achieving a Mean Absolute Percentage Error (MAPE) of 6.763%, a Mean Absolute Error (MAE) of USD 3.935 billion, a Root Mean Squared Error (RMSE) of USD 6.850 billion and an R^2 value of approximately 0.990. XGBoost and Histogram-Based Gradient Boosting also achieved strong results, while the Seasonal LSTM produced a higher MAPE of 17.214%.

Model explainability was assessed using SHAP analysis, which showed that the exact 12-month historical trade lag was the most influential predictor in the final model. Among the external variables, base-metals index had the greatest influence on predictions. Including external indicators reduced MAPE from 8.225% to 6.763% and reduced MAE from USD 4.032 billion to USD 3.935 billion. However, RMSE increased from USD 6.473 billion to USD 6.850 billion and R^2 decreased slightly from 0.992 to 0.990. The Random Forest tree-percentile ranges achieved empirical coverage of 93.519%, although these ranges were not formally calibrated.

Overall, the results show that historical seasonal trade behaviour provided the strongest forecasting signal, while external indicators added limited and uneven predictive value.

1 Introduction

1.1 Background

International trade supports economic activity by enabling goods and services to move across national borders [2]. Accurate trade forecasts can help governments in monitoring economic performance, support businesses with production and supply-chain planning, and help financial institutions assess trade-finance requirements and related risks [3]. However, monthly trade values can change because of seasonality, differences between national economies and changes in global economic conditions

Historical trade values provide an important basis for forecasting because import and export activity often follows recurring annual patterns. External economic indicators may also provide useful information [4]. Commodity prices can influence the value and cost of traded goods, while interest rates, inflation, exchange-rate conditions and industrial activity can influence demand, investment and competitiveness. However, it is not always clear whether these external indicators improve forecasting accuracy beyond the information already contained in historical seasonal trade values.

This study therefore compares machine-learning models for forecasting monthly import and export values across 18 G20 economies. The main research question is whether oil prices, commodity indices and selected macroeconomic indicators improve forecasting accuracy beyond the information contained in an exact 12-month historical trade lag.

1.2 Aims

The aim of this study is to develop, evaluate and compare machine learning models for forecasting monthly import and export values across selected G20 economies using historical trade patterns and external economic indicators.

To achieve this aim, the study has the following objectives:

- to collect, clean and integrate monthly trade data and external economic indicators
- to construct exact historical lags, seasonal variables and lagged external features
- to develop and compare statistical, ensemble, boosting and deep-learning forecasting models

- to evaluate all models using a chronological 2024 holdout period and standard regression metrics
- to compare a seasonal history-only model with a model that also includes external indicators
- to analyse model performance and forecasting errors at the country level
- to examine model explainability, uncertainty and sensitivity using SHAP, tree-percentile ranges, ablation analysis and scenario analysis

1.3 Technologies

Python was used throughout the project for data preparation, feature engineering, model development, evaluation and visualisation. The main forecasting notebook was developed in Jupyter Notebook through the Anaconda distribution on a MacBook. **Pandas** and **NumPy** were used for data manipulation and numerical operations.

Scikit-learn was used to implement Linear Regression, Support Vector Regression, Random Forest and Histogram-Based Gradient Boosting models.

XGBoost was used for gradient boosting models, while **TensorFlow/Keras** was used to develop and evaluate the Long Short-Term Memory (LSTM) neural network [5] separately in Google Colab. **SHAP** was applied to explain model predictions by quantifying the contribution of individual features [6]. **Matplotlib** was used to visualise data distributions, model performance and feature importance.

Monthly trade data were obtained through the **UN Comtrade API**, while external economic indicators, including commodity prices, oil prices and macroeconomic variables, were collected from the **Federal Reserve Economic Data (FRED)** database and other publicly available economic data sources before being integrated into the forecasting pipeline.

2 System

2.1 Requirements

The project requirements developed as the datasets were explored and the forecasting pipeline was developed. The initial requirement was to create a machine-learning model capable of forecasting monthly trade values. As the project progressed, this expanded to include reliable data collection, accurate alignment of monthly datasets, construction of exact lag features, comparison of several forecasting models, country-level performance analysis, model explainability and an assessment of prediction uncertainty.

2.1.1 Functional Requirements

Following the requirements analysis phase, the functional requirements were reviewed and updated to reflect the final system design, implementation and research objectives. Table 1 below summarises the final functional requirements implemented in this project.

Table 1 - Functional requirements

ID	User Story	Priority	Status (Delivered / Partial / Not delivered)
FR-001	As a student researcher, I want to collect monthly import and export data for selected G20 economies so that a historical forecasting dataset can be created.	MUST	Delivered
FR-002	As a student researcher, I want to integrate oil prices, commodity indices and macroeconomic variables so that external economic conditions can be incorporated into the forecasting models.	MUST	Delivered
FR-003	As a student researcher, I want to clean and preprocess the combined datasets so that they are suitable for machine learning.	MUST	Delivered
FR-004	As a student researcher, I want to generate lagged and seasonal features so that historical trade patterns can be captured by the forecasting models.	MUST	Delivered
FR-005	As a student researcher, I want to train and compare multiple machine learning models so that the most accurate forecasting approach can be identified.	MUST	Delivered
FR-006	As a student researcher, I want to evaluate forecasting performance using standard regression metrics so that model accuracy can be objectively assessed.	MUST	Delivered
FR-007	As a student researcher, I want to compare history-only models with models incorporating external economic indicators so that the contribution of external variables can be evaluated	MUST	Delivered
FR-008	As a student researcher, I want to analyse forecasting performance at both the overall and individual country level so that model strengths and weaknesses can be identified	SHOULD	Delivered
FR-009	As a student researcher, I want to explain model predictions using SHAP so that the influence of each feature can be understood.	SHOULD	Delivered
FR-010	As a student researcher, I want to estimate prediction uncertainty and perform ablation and scenario analysis so that model robustness can be evaluated.	SHOULD	Partial

2.1.2 Non-Functional Requirements

The non-functional requirements focused on performance, reproducibility, extensibility and the clarity of the final outputs. Where execution time or retrieval rates were not formally recorded, the requirement is reported as not formally measured rather than assumed to have been achieved.

Table 2 - Non-Functional Requirements

ID	Requirement (with measurable target)	Priority	Target met?
NFR-001	The complete data preprocessing pipeline shall execute within 15 minutes on the development environment.	SHOULD	Not formally measured

ID	Requirement (with measurable target)	Priority	Target met?
NFR-002	Traditional machine learning models shall complete training within 10 minutes, and the LSTM model within 60 minutes	SHOULD	Not formally measured
NFR-003	The data acquisition pipeline shall successfully retrieve and process at least 95% of requested data while handling missing values and API interruptions.	MUST	Partially evidenced; resumable downloading was implemented, but the final retrieval percentage was not formally calculated
NFR-004	The best-performing forecasting model shall achieve MAPE below 7% and R^2 above 0.95 on the chronological test dataset.	MUST	Met (MAPE = 6.763%, R^2 = 0.990)
NFR-005	The forecasting pipeline shall produce reproducible results using the same datasets, parameters and random seeds, with evaluation metrics remaining within ±2% variation	MUST	Met for the final Random Forest
NFR-006	The solution shall maintain a modular architecture, separating data acquisition, preprocessing, feature engineering, modelling and evaluation into independent components	SHOULD	Partially met
NFR-007	The pipeline shall support the addition of new countries and external economic indicators without requiring significant redesign.	SHOULD	Partially Met
NFR-008	The system shall present model outputs using clear evaluation tables, comparison charts and explainability visualisations to support interpretation of forecasting results.	SHOULD	Met

2.1.3 Data Requirements (AI / ML and data-driven projects)

The dataset was prepared to support the development and evaluation of machine-learning models for forecasting monthly international trade activity. Monthly import and export values were collected from the UN Comtrade API. Macroeconomic indicators were obtained mainly from the Federal Reserve Economic Data database [7], while commodity-price series were obtained from World Bank Commodity Price Data [8], with other publicly available economic sources.

The study included 18 G20 economies. Russia was excluded because it was not part of the final dataset collected for the project. For each country, monthly import and export records were retained as separate observations. The original data covered the period from January 2014 to December 2024. After data cleaning, monthly alignment and the creation of an exact 12-month trade lag, the final modelling dataset contained 4,196 observations.

The target variable was monthly primaryValue, representing either the import or export value for a country in a particular month. Before model training, the target was transformed using log1p to reduce skewness and limit the effect of extremely large trade values. After the models generated their predictions, the results were converted back into U.S. dollars using the corresponding expm1 transformation before the evaluation metrics were calculated.

The final controlled model comparison used 35 predictor variables. These included month, quarter, the exact 12-month trade lag, 14 external indicators lagged by one month, 17 country dummy variables and one trade-flow dummy variable. The external indicators covered crude-oil prices, energy and non-energy indices, food, fertiliser, base-metal and precious-metal indices, the federal funds rate, the broad U.S. dollar index, U.S. industrial production, U.S. CPI, the U.S. 10-year Treasury rate, global policy uncertainty and a deep-sea freight producer-price index.

A chronological split was applied to reflect a realistic forecasting situation. The seasonal dataset contained 3,764 training observations covering 2015 to 2023 and 432 test observations covering January to December 2024. The 2024 records were excluded from the training of the traditional machine-learning models. For the final LSTM experiment, target sequences were used for validation and epoch selection, and the final model was then retrained using all target sequences up to the end of 2023 before being evaluated on the 2024 test period.

2.2 Design and Architecture

The forecasting system was designed as a sequential and reproducible data pipeline [9]. Each stage produced an output that was passed to the next stage, allowing data collection, preparation, feature engineering, model development and evaluation to be examined separately. Figure 1 below presents the complete workflow, beginning with data acquisition and ending with the final interpretability and robustness analyses.

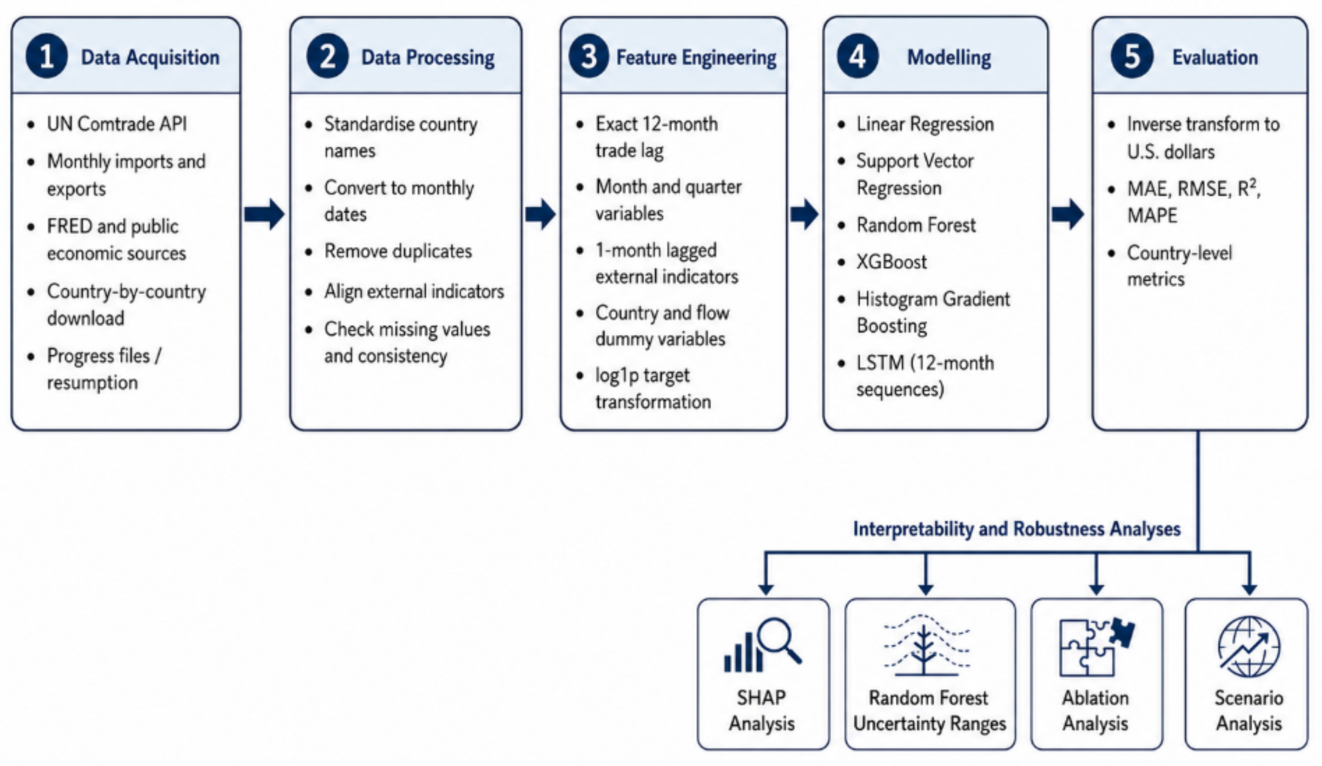

Figure 1 - Complete Forecasting Workflow

Data Acquisition Layer: The data-acquisition layer collected monthly import and export data from the UN Comtrade API together with external economic indicators from FRED and other publicly available sources. Trade data were downloaded separately for each country and reporting month.

Data Processing Layer: The processing layer standardised country names, converted reporting periods into a consistent monthly date format, removed duplicate records and aligned the external indicators with the corresponding trade observations. Import and export values were retained as separate records.

Feature Engineering Layer: The feature-engineering layer created the exact 12-month historical trade lag, calendar variables and external indicators lagged by one month. Country and trade-flow categories were converted into dummy variables. The target variable was transformed using log1p, and the final predictor set was checked to confirm that the training and test datasets contained the same columns and did not include any missing values.

Modelling Layer: The modelling layer trained Linear Regression, Support Vector Regression, Random Forest, XGBoost and Histogram-Based Gradient Boosting using the same 35 predictors and the same chronological training dataset. The LSTM model was evaluated separately using 12-month sequences containing the same 35 predictor variables. Fixed random seeds were applied to the models where this was supported.

Evaluation Layer: Model predictions were converted back into U.S. Dollars and evaluated using MAE, RMSE, R^2 and MAPE. Further evaluation included country-level performance metrics, SHAP analysis, Random Forest tree-percentile uncertainty ranges, selected-feature ablation and scenario analysis.

2.3 Implementation

2.3.1 Data Collection

Monthly international trade data were collected from the UN Comtrade API using the relevant reporter code for each selected country. Each country was processed separately instead of requesting the complete dataset in a single API call. This reduced the impact of API interruptions and made it easier to identify and correct any incomplete country downloads.

For every reporting country, monthly import and export records were collected for the period from January 2014 to December 2024. The successfully downloaded observations were added to separate country-level files. Progress was saved after each completed month so that, if an interruption occurred, the download could continue from the most recent completed period rather than restarting the full process.

Oil prices, commodity indices and macroeconomic indicators were collected separately from publicly available economic databases. These datasets were stored in separate source files until the data-cleaning and monthly alignment stages had been completed.

2.3.2 Data Cleaning and Integration

Country names and date fields were standardised across the trade and external datasets before they were merged. Monthly reporting periods were converted into a consistent first-of-month date format. Duplicate trade records were removed, while import and export values were retained as separate observations using the relevant trade-flow code.

External variables were aligned with the trade data by reporting month. Before model development, each external variable was lagged by one month so that the value used for prediction represented information available from the previous month. The datasets were then combined into a single pooled structure containing observations for all 18 countries and both trade flows.

The integrated dataset was checked for missing values, duplicate country-flow-month records and inconsistent categories. Rows without the information required for the final feature set were removed before the modelling dataset was prepared.

2.3.3 Feature Engineering

The main historical feature used in the study was the exact 12-month trade lag. For each country, trade flow and target month, the corresponding observation from exactly one calendar year earlier was identified using the reporting date. This date-based approach prevented an observation from being treated as a 12-month lag where a monthly record was missing or the sequence was incomplete. A validation check was also carried out to confirm that every matched lag date was exactly one year before the target date.

Month and quarter were included as calendar variables, while the 14 external indicators were included using their values from the previous month. Country names were represented using 17 dummy variables, and export flow was represented by one dummy variable, with the excluded categories used as the reference groups. The year variable was not included in the final controlled model comparison.

The monthly trade target was transformed using log1p before model training. The final controlled predictor dataset contained 35 features, with 3,764 observations used for training and 432 observations used for testing. The training and test datasets were checked to confirm that their columns were identical and in the same order, and that no missing predictor values remained.

2.3.4 Model Training

Six forecasting approaches were evaluated. Linear Regression was used as the statistical baseline, while Support Vector Regression applied a radial basis function kernel with C set to 10 and epsilon set to 0.1 [10]. Both models were implemented using pipelines that standardised the predictors before training.

The final Random Forest used 200 decision trees, random_state=42 and parallel processing through n_jobs=-1 [11]. XGBoost used 300 estimators, a maximum depth of four, a learning rate of 0.05 and the squared-error regression objective [12]. Histogram-Based Gradient Boosting used 300 iterations, a maximum depth of four, a learning rate of 0.05 and random_state=42 [13].

The final LSTM used 12-month input sequences with 35 predictors at each time step [14]. The network included a 64-unit LSTM layer, dropout of 0.2 [15], a 32-unit LSTM layer, a second dropout layer, a 16-unit dense layer with ReLU activation and a one-unit output layer. The model contained 38,561 trainable parameters. The predictors and target values were standardised using the training data only.

For epoch selection, target sequences up to 2022 were used for training, while 2023 sequences were used for validation. Early stopping identified epoch 12 as the point with the lowest validation loss [16]. The final LSTM was then rebuilt and trained for 12 epochs using all target sequences up to 2023 before evaluation on the 432 target sequences from 2024.

2.3.5 Prediction and Inverse Transformation

All models predicted the logarithmically transformed target. After prediction, the values were converted back to U.S. dollars using the inverse of the log1p transformation:

$$\hat{y} = \exp(\hat{y}_{log}) - 1$$

The inverse transformation was applied before calculation of MAE, RMSE, R^2 and MAPE. MAE, RMSE and MAPE are commonly used measures of forecast accuracy [17]. Country-level metrics, residuals, tables and figures were therefore reported using actual monthly trade values rather than logarithmic units.

2.3.6 Explainability and Uncertainty

SHAP was applied to the final Random Forest model using TreeExplainer [18]. SHAP values were calculated for all 432 observations in the 2024 test dataset. The global analysis ranked the features using their mean absolute SHAP values, and a SHAP summary plot was created to show the magnitude and direction of the feature contributions. The SHAP results explain how individual features contributed to the model's predictions, but they do not establish causal economic relationships.

Prediction uncertainty was estimated from the distribution of predictions generated by the 200 individual Random Forest trees. The approach was informed by quantile-based Random Forest methods [19]. For each test observation, the 2.5^{th} and 97.5^{th} percentiles were used to create lower and upper model-based ranges, which were then compared with actual values to calculate empirical coverage. However, these ranges were calculated directly from the individual-tree predictions and were not formally calibrated as statistical prediction intervals or confidence intervals [20].

2.3.7 Ablation and Scenario Analysis

Ablation analysis was carried out to assess the contribution of selected external variables [20]. One variable was removed at a time, the Random Forest model was retrained using the remaining predictors, and the revised 2024 MAPE was compared with the complete model's MAPE of 6.763%. The variables tested were the base-metals index, federal funds rate, broad U.S. dollar index, U.S. CPI, non-energy index and crude-oil prices.

Scenario analysis was carried out by changing one selected feature at a time while keeping all other predictor values unchanged. Three scenarios were tested: a 20% increase in crude-oil prices for Saudi Arabia, a 15% decrease in the base-metals index for Australia and a one-percentage-point increase in the federal funds rate across the complete test dataset. The resulting forecast changes show how sensitive the trained model was to these variables and were not interpreted as causal estimates.

2.4 Model Validation

Model validation was carried out to ensure that the forecasting pipeline produced reliable, consistent and reproducible results. Validation focused on maintaining a clear separation between training and testing data, confirming that the engineered features were created correctly and ensuring that all models were evaluated using the same experimental conditions.

A chronological train-test split was used instead of a random split to reflect a realistic forecasting scenario [17]. The final seasonal dataset contained 3,764 observations from 2015 to December 2023 for training, while 432 observations from January to December 2024 were reserved exclusively for testing. This ensured that the reported evaluation metrics were calculated using future observations that had not been included during the training of the traditional machine-learning models.

Preventing information leakage was an important part of the validation process. The 12-month trade lag was created by matching each target observation with the same country, trade flow and month exactly one calendar year earlier. The matched dates were checked to confirm that every historical value was exactly 12 months before the corresponding target observation. External economic indicators were also incorporated using one-month lagged values so that the models only used information that would have been available before each forecast was generated.

Before model training, the final predictor dataset was checked for missing values and inconsistencies between the training and test data. The year variable was excluded from the final controlled model comparison, while the training and test feature matrices were checked to confirm that they contained the same 35 predictors in the same order. No missing predictor values remained in either dataset.

The log1p transformation of the target variable and the corresponding expm1 inverse transformation were tested to ensure that generated predictions could be accurately returned to their original monetary values. The same 35 predictors, 3,764 training observations and 432 test observations were used across all traditional machine-learning models. This ensured that differences in forecasting performance reflected the selected models rather than changes in the feature sets or evaluation data.

Fixed random seeds were applied to the Random Forest, XGBoost, Histogram-Based Gradient Boosting and LSTM experiments where supported. The Random Forest model was also trained twice using the same configuration. Both runs produced the same MAPE of 6.763%. The maximum difference between corresponding predictions was less than USD 0.01 and the predictions were reproducible within the defined numerical tolerance. This confirmed that the final Random Forest results were reproducible under the selected experimental settings.

2.5 Experimental Configuration

Table 3 below summarises the software environments, data split, main libraries and hardware used during the experiments.

Table 3 - Experimental Configuration

Component	Configuration
Operating System	macOS
Development Environment	Jupyter Notebook through Anaconda
Python Version	Python 3.13.9
LSTM environment	Google Colab
Final Controlled Training Data	3,764 observations
Final controlled test data	432 observations from 2024
Primary Libraries	Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow/Keras, SHAP, Matplotlib
Hardware	Apple MacBook (Apple M4, 24GB)

2.6 Experimental Procedure

The experiment was carried out using a fixed sequence of steps. Monthly import and export data and the external economic indicators were first collected and stored separately. Country names and reporting dates were standardised, duplicate records were removed and the datasets were aligned using a common monthly date. The exact 12-month trade lag, one-month-lagged external variables, calendar features and dummy variables were then created. The target variable was transformed using log1p before the chronological training and test datasets were prepared.

The traditional machine-learning models were trained using the same 35 predictors and evaluated on the 2024 test dataset after their forecasts had been converted back to the original monetary values. The LSTM model used 12-month predictor sequences and was evaluated using the same 2024 target period. The strongest-performing traditional model was then examined further using country-level performance metrics, SHAP analysis, model-based uncertainty ranges, ablation analysis and scenario analysis.

2.7 Results

2.7.1 Overall Model Performance

Table 4 below summarises the overall forecasting performance achieved by each model using the same 2024 holdout period. Seasonal LSTM used the same 2024 target period but was trained through a separate 12-month sequence workflow.

Table 4 - Forecasting performance achieved by each model

Model	MAE USD bn	RMSE USD bn	R^2	MAPE (%)
Linear Regression	13.417	25.980	0.863	16.039
Support Vector Regression	10.886	20.915	0.911	20.704
Random Forest	3.935	6.850	0.990	6.763
Histogram-Based Gradient Boosting	4.084	7.000	0.990	7.219

XGBoost	4.165	7.420	0.989	7.057
Seasonal LSTM	13.169	32.603	0.784	17.214

The Random Forest achieved the lowest MAE, RMSE and MAPE, together with an R^2 of 0.990, making it the strongest overall model. XGBoost and Histogram-Based Gradient Boosting also produced strong results, with MAPE values of 7.057% and 7.219% respectively. The relatively small difference between the three tree-based models suggests that the final lagged and seasonal feature set was well suited to non-linear ensemble methods.

Linear Regression produced a higher MAPE of 16.039%, indicating that a single linear relationship was not able to capture the combined effects of country differences, seasonal patterns and external economic conditions effectively. The Seasonal LSTM achieved a MAPE of 17.214% and a lower R^2 of 0.784. Support Vector Regression recorded the highest MAPE at 20.704%, although its MAE was lower than the values produced by Linear Regression and the LSTM. Overall, the results supported the selection of the Random Forest model for the detailed country-level, explainability and uncertainty analyses.

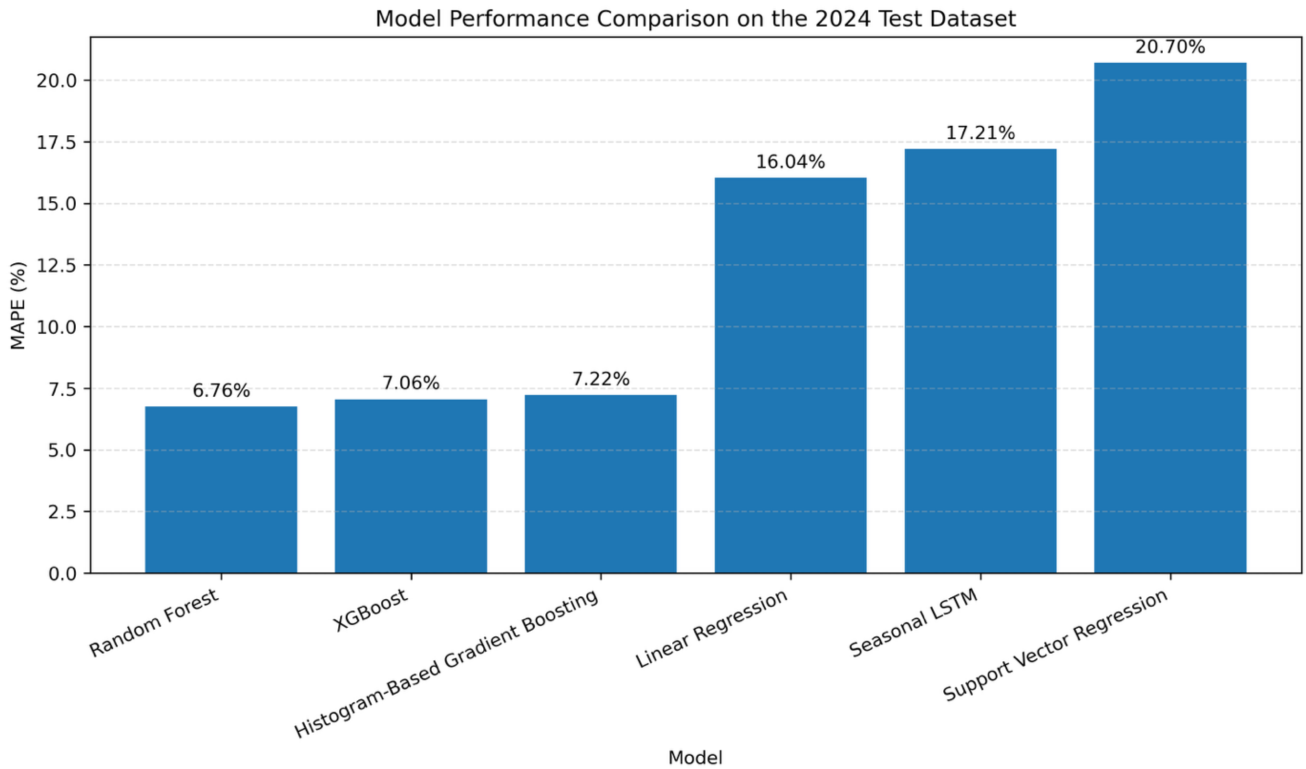

Figure 2 - Model Performance comparison on the 2024 Test Data

2.7.2 Country-Level Performance

To evaluate the consistency of the forecasting framework, performance was also analysed separately for each reporting country.

Country-level forecasting accuracy varied across the 18 economies. France achieved the lowest MAPE at 4.20%, followed by Canada at 4.40% and Japan at 4.50%. Argentina recorded the highest MAPE at 16.04%. However, MAPE and R^2 did not always provide the same view of model performance. Canada produced a relatively low percentage error but recorded an R^2 of -1.01, while the Republic of Korea recorded an R^2 of -2.26. These negative R^2 values indicate that, for these countries, the model was less effective at explaining monthly variation than a simple benchmark based on the mean value.

China and the United States produced the largest absolute errors because their monthly trade values were substantially higher than those of most other countries. Their MAE values were approximately USD 12.82 billion and USD 12.57 billion respectively. Despite this, their MAPE values remained below 5%, while their R^2 values were approximately 0.87 and 0.90. Overall, the pooled model estimated general trade levels effectively for many countries, but its ability to capture short-term monthly variation was not consistent across all economies.

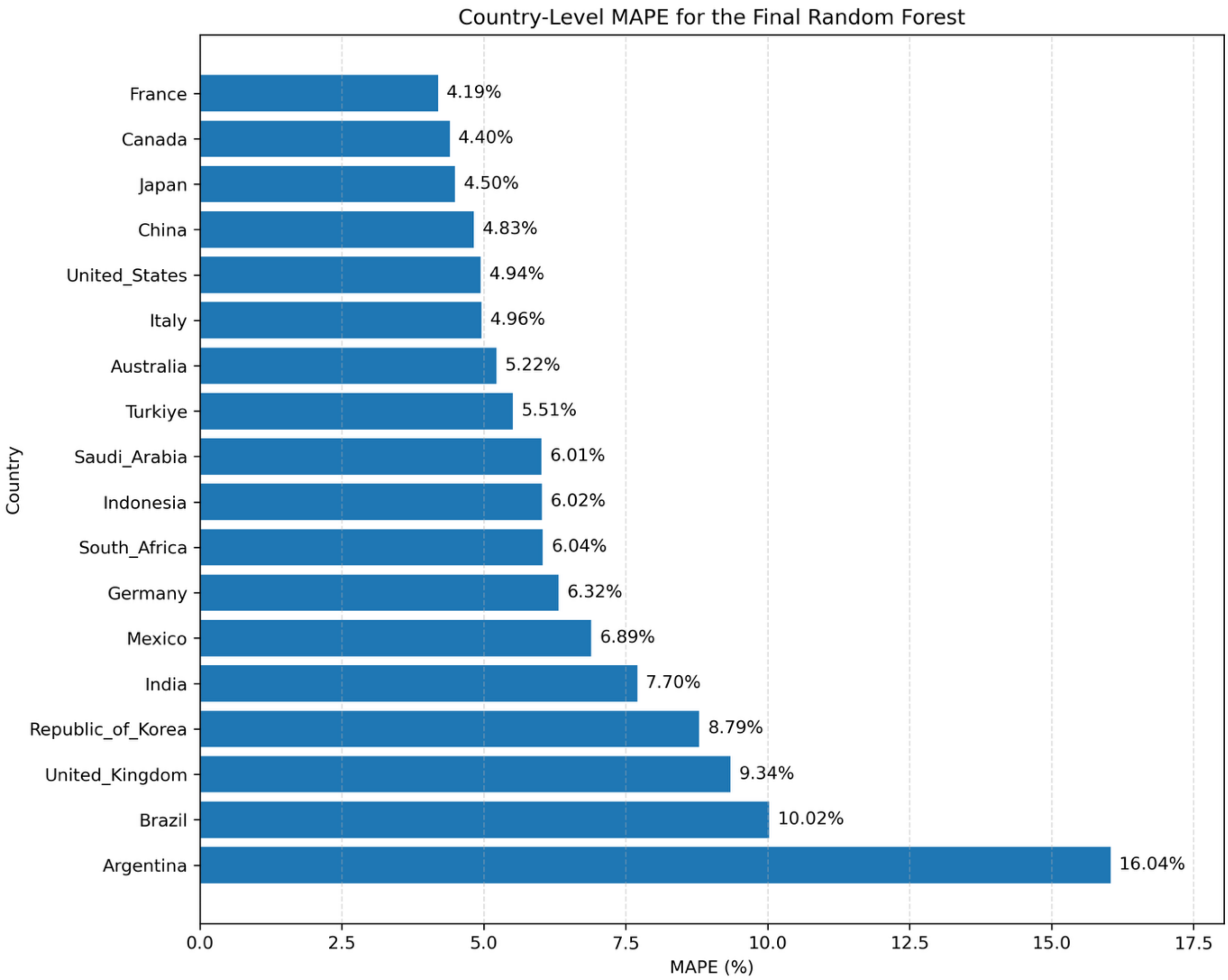

Figure 3 - Country-level MAPE for final Random Forest on the 2024 test dataset

2.7.3. History-Only versus External Variables

The study compared two Seasonal Random Forest configurations to assess whether external economic indicators improved forecasting beyond historical trade information. Table 5 presents the results for the history-only model and the model incorporating oil prices, commodity indices and macroeconomic variables.

Table 5 - Forecasting Accuracy – Seasonal History only vs Seasonal History with External Indicators

Model	MAE USD bn	RMSE USD bn	R^2	MAPE
Seasonal history only	4.032	6.473	0.992	8.225
Seasonal history plus external indicators	3.935	6.850	0.990	6.763

Adding external economic indicators reduced the overall MAPE from 8.225% to 6.763%, representing an improvement of 1.462 percentage points. MAE also decreased from USD 4.032 billion to USD 3.935 billion. However, RMSE increased from USD 6.473 billion to USD 6.850 billion and R^2 decreased slightly from 0.992 to 0.990.

The results therefore show that the external indicators improved both average absolute error and relative percentage accuracy, but did not improve every evaluation metric. Historical seasonal trade information remained the strongest source of predictive value, while the contribution of the external indicators was modest and depended on the metric being considered.

2.7.4 Feature Importance

SHAP analysis was carried out on the final Random Forest model to examine the contribution of each predictor to the forecast generated for 2024. The exact 12-month trade lag was identified as the most influential feature by a considerable margin. This confirms that the trade value recorded for the same country, trade flow and month in the previous year provided the strongest predictive information within the model.

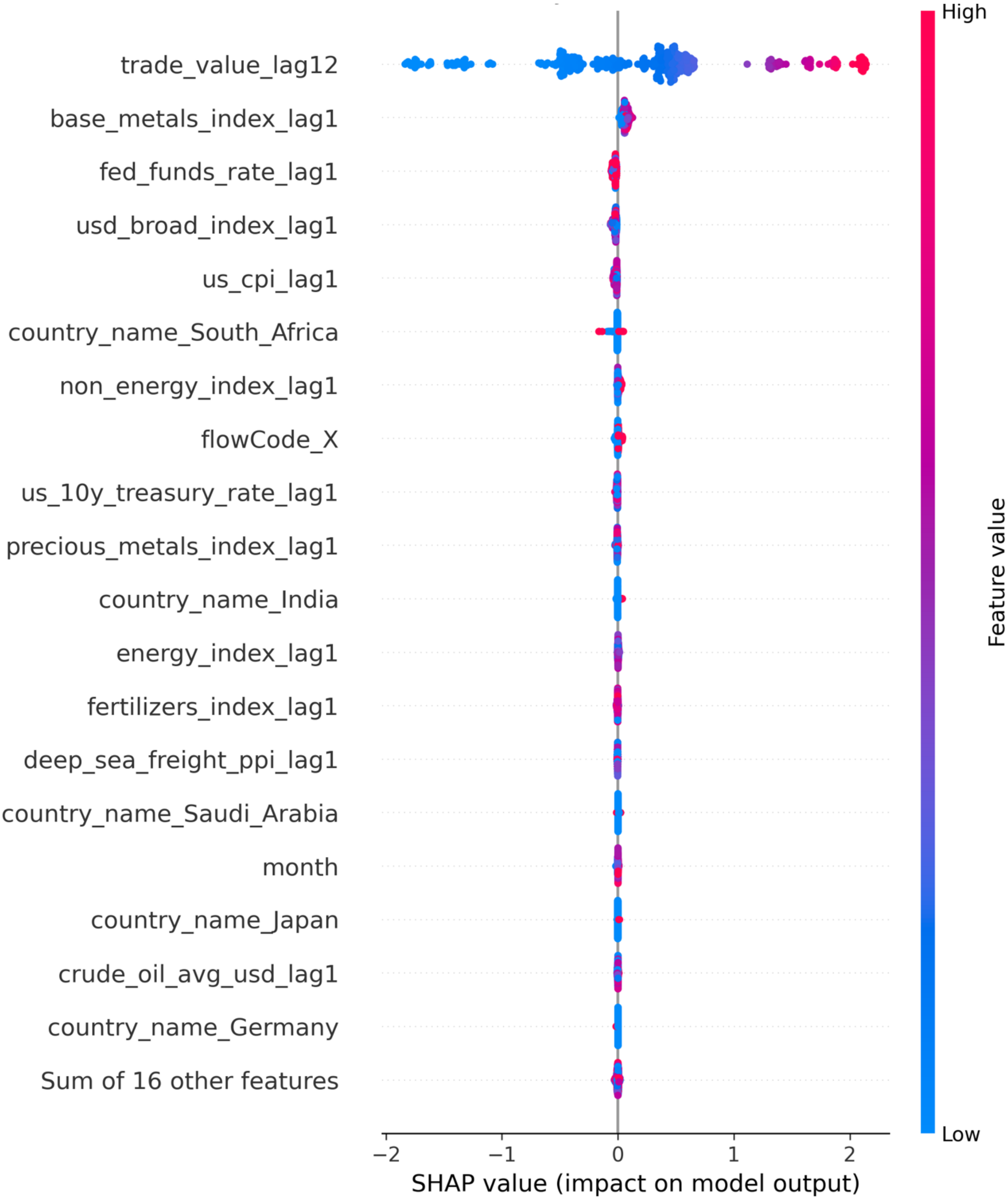

Figure 4 - SHAP summary showing feature importance

Among the external economic indicators, the base-metals index produced the highest mean absolute SHAP value. The federal funds rate and the broad U.S. dollar index were the next most influential external variables, followed by U.S. CPI, the non-energy index and the U.S. 10-year Treasury rate. Crude-oil prices made a comparatively smaller contribution to the model predictions. Overall, the findings show that the model relied mainly on historical seasonal trade patterns, while the external indicators provided smaller adjustments to individual forecasts.

The SHAP values explain how the trained model used each feature when generating its predictions. However, these values should not be interpreted as evidence that changes in the external indicators directly caused corresponding changes in international trade.

2.7.5 Prediction Uncertainty

Prediction uncertainty was estimated using the forecasts generated by the 200 individual trees within the Random Forest model. For each observation in the test dataset, the 2.5th and 97.5th percentiles of the tree predictions were used to create lower and upper model-based ranges.

Across the 432 test observations, 93.519% of the actual trade values fell within the tree-percentile ranges. The mean interval width was USD 19.626 billion, while the median interval width was USD 15.887 billion. Coverage also varied between countries. The United States, China and India recorded lower coverage of 83.33%. The Republic of Korea, Brazil, the United Kingdom and Germany recorded coverage of 87.50%, while Mexico and Argentina recorded coverage of 91.67%. The remaining countries achieved coverage of 100%.

The high overall coverage shows that the estimated ranges included most of the observed trade values. However, the method was not formally calibrated. The ranges represent the variation between the fitted Random Forest trees and should not be interpreted as statistical confidence intervals.

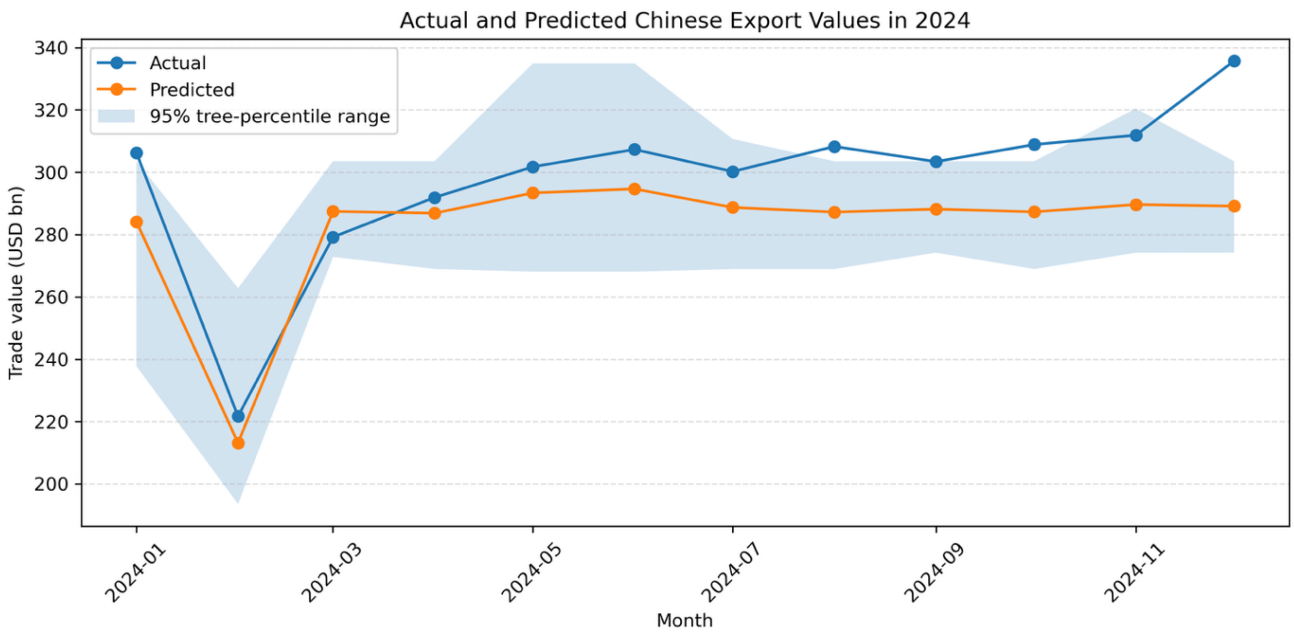

Figure 5 - Actual and Predicted monthly Chinese Export values for 2024 with Random Forest tree-percentile uncertainty ranges

2.7.6 Ablation and Scenario Analysis

Ablation analysis was carried out by removing one external indicator, retraining the Random Forest and comparing the resulting MAPE with the complete model value of 6.763%.

Table 6 - Ablation analysis outcome after removal of each selected external indicator

Removed feature	Revised MAPE (%)	Change (percentage points)
Base-metals index	6.969	+0.206
Broad U.S. dollar index	6.829	+0.066
Non-Energy index	6.826	+0.063
Federal funds rate	6.810	+0.047
Crude-oil price	6.584	-0.179
U.S. CPI	6.546	-0.217

Ablation analysis showed that removing the base-metals index produced the largest reduction in forecasting accuracy, increasing MAPE by 0.206 percentage points. Removing the broad U.S. dollar index, non-energy index and federal funds rate also increased MAPE, although their effects were smaller. In contrast, removing crude-oil prices and U.S. CPI reduced MAPE by 0.179 and 0.217 percentage points respectively. Under the selected model configuration, these two variables therefore did not provide a positive contribution to out of sample percentage accuracy. These results relate to the trained forecasting model and should not be interpreted as evidence of causal economic relationships.

Three scenario tests were also carried out. A 20% increase in crude-oil prices increased the Saudi Arabian forecasts by approximately 0.447% on average, with individual changes ranging from -0.174% to 1.315%. A 15% reduction in the base-metals index reduced the Australian forecasts by approximately 2.197% on average, with individual changes ranging from -4.871% to 0.699%. Increasing the federal funds rate by one percentage point produced an average forecast change of approximately -0.0069% across the complete test dataset, with individual changes ranging from -0.348% to 0.241%.

These experiments show how the trained model responded when selected input values were changed. However, they do not demonstrate causal economic relationships and should not be interpreted as direct estimates of how future trade would respond to actual policy or market changes.

2.8 Evaluation

2.8.1 Achievement of Research Objectives

The main forecasting objectives of the project were achieved. Monthly trade data and external indicators were collected, cleaned and combined into a single modelling dataset. Exact historical trade lags and lagged economic features were created, several forecasting models were implemented and the final models were evaluated using the 2024 chronological holdout period. Through this controlled comparison, the Random Forest was identified as the strongest overall model.

The robustness objective was partially achieved. Selected-feature ablation and scenario analyses were completed, but the uncertainty ranges were not formally calibrated and the scenario tests covered only a limited number of model-sensitivity experiments.

2.8.2 Discussion of Model Performance

The final results show that non-linear tree-based models were better suited to the pooled international trade dataset than the linear, kernel-based and recurrent models. The Random Forest achieved the lowest MAE, RMSE and MAPE, while XGBoost and Histogram-Based Gradient Boosting produced closely comparable results. This indicates that interactions between the historical trade lag, country, trade flow, seasonal features and external indicators were important for forecasting performance.

Linear Regression was less effective at capturing these relationships. Support Vector Regression produced a lower MAE than Linear Regression but recorded the highest MAPE, indicating that its percentage accuracy was uneven across observations with different trade values. The LSTM also performed less effectively than the tree-based models.

Although the LSTM sequence dataset contained several thousand pooled samples, each country and trade-flow series contained only approximately nine years of usable monthly target observations after sequence construction. The countries may also have followed different sequential patterns, which reduced the ability of one pooled recurrent model to learn a common structure.

The LSTM result should therefore not be interpreted as evidence that deep learning is unsuitable for international trade forecasting. It only shows that the selected architecture, the available monthly history and the pooled sequence design did not outperform the tree-based models in this study. More extensive tuning, longer time series or separate country-level sequence models may produce different results.

2.8.3 Contribution of External Economic Indicators

The external-indicator experiment produced mixed results. Adding the 14 lagged external variables reduced the overall MAPE by 1.462 percentage points and reduced MAE by USD 0.097 billion. However, RMSE increased by USD 0.377 billion and R^2 decreased slightly by 0.002. This means that the external indicators improved average absolute and relative percentage accuracy but did not improve every measure of model performance.

Historical seasonal trade behaviour remained the strongest source of predictive information within the model. SHAP and ablation analysis identified the base-metals index as the external variable with the clearest positive contribution. The broad U.S. dollar index, non-energy index and federal funds rate also provided smaller improvements. In contrast, removing crude-oil prices and U.S. CPI improved MAPE. The external indicators therefore provided additional information, but their contribution differed across evaluation metrics and individual variables.

2.8.4 Model Interpretability and Limitations

One of the strengths of the final forecasting framework is that the Random Forest combined strong test performance with interpretable model analysis. SHAP showed how individual features contributed to the forecasts, while the ablation and scenario analyses provided further information about the model's sensitivity to selected variables. The tree-percentile ranges also allowed uncertainty to be presented alongside the point forecasts rather than reporting a single prediction without additional context.

Despite these strengths, several limitations remain. First, the study used one pooled model across 18 countries and both import and export flows. This increased the number of available observations but may have reduced the model's ability to capture country-specific and trade-flow relationships. Second, model performance was evaluated using only one final holdout year. A rolling-origin or expanding-window evaluation covering several years would provide stronger evidence of long-term generalisability.

Country-level performance was also inconsistent. Some countries achieved low MAPE values but negative R^2 values, showing that the model estimated the overall level of trade more effectively than the month-to-month variation for those countries.

The external indicators were mainly global or U.S.- centred and may not have represented the domestic economic conditions of every economy included in the study. Geopolitical events, trade-policy changes and supply-chain disruptions were also not modelled directly. In addition, the Random Forest uncertainty ranges were not formally calibrated, while the scenario analysis measured the sensitivity of the trained model rather than the causal economic effects.

3 Conclusions

This study developed and evaluated a machine-learning framework to forecast monthly import and export values across 18 G20 economies. Historical trade data were combined with selected commodity-price and macroeconomic indicators, and the models were tested using a chronological test period covering January to December 2024.

The Random Forest model using seasonal and external predictors produced the strongest overall results. It achieved an MAE of USD 3.935 billion, an RMSE of USD 6.850 billion, and R^2 of 0.990 and a MAPE of 6.763%. The exact 12-month trade lag was the most influential feature, showing that recurring annual trade patterns provided the strongest forecasting information. XGBoost and Histogram-Based Gradient Boosting also performed well, while Linear Regression, Support Vector Regression and the Seasonal LSTM produced higher errors.

Adding external indicators reduced MAPE from 8.225% to 6.763% and reduced MAE from USD 4.032 billion to USD 3.935 billion. However, RMSE increased slightly and R^2 decreased from 0.992 to 0.990. The base-metals index made the clearest positive external contribution, while crude-oil prices and U.S. CPI did not improve MAPE under the selected model configuration.

SHAP, uncertainty, ablation and scenario analysis were used to support interpretation of the results. The uncertainty ranges achieved empirical coverage of 93.519%, although they were not formally calibrated. Overall, the findings show that pooled Random Forest using seasonal trade history and selected lagged external indicators can produce accurate and interpretable forecasts.

4 Further Development or Research

Future research should first strengthen the model-validation process. Rolling-origin or expanding-window testing across several years would provide a clearer indication of whether the forecasting model remains reliable under different economic conditions. This would provide a more robust evaluation than relying on a single chronological holdout period covering 2024.

Country-specific models, separate import and export models and hierarchical approaches combining pooled and local information could also be compared with the current global model. These approaches may improve forecasting performance for countries that recorded negative R^2 values. Formally calibrated prediction intervals should also be developed so that forecast uncertainty can be interpreted more reliably.

The predictor dataset could be expanded to include additional country-specific macroeconomic indicators and more timely measures of shipping activity, freight rates, purchasing manager's indices, trade-policy changes and supply-chain disruption. Event-based variables could also be introduced to represent major geopolitical developments or economic shocks that may not be captured by historical trade patterns or monthly economic indicators.

Alternative sequence and transformer-based models could be examined after the validation framework and country-level modelling have been strengthened [22]. The forecasting pipeline could also be developed into an operational system by automating data collection, model retraining and forecast generation. This could provide regularly updated trade forecasts to support economic analysis, trade-finance decisions and supply-chain planning.

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6 Appendix

6.1 Project Proposal

Project Proposal

Team Project Module · National College of Ireland · Individual Project

Project Name: Forecasting International Trade Activity using Oil Prices and Macroeconomic Indicators: A machine learning analysis of G20 Economies

Student Name: John Selvin Raja John
Student Number: 25120051
Module / Cohort: HDAIML_SEP25OL

Date: 8 June 2026

1. Objectives & Problem

Global Trade is highly influenced by fluctuations in oil prices due to their impact on transportation costs, production expenses, supply chain operations and overall economic activity. Changes in oil prices can affect the movement of goods across international markets, influencing both import and export volumes. However, the relationship between oil prices and trade activity is complex, dynamic and often differs across countries.

The objective of this project is to investigate whether oil price movements can be used to predict international trade activity across G20 economies. Specifically, the project aims to develop machine learning models capable of forecasting monthly import and export values using historical trade data, oil prices and macroeconomic indicators. The expected outcome is a predictive framework that provides insights into how energy market fluctuations influence international trade performance and supports more informed business and economic decision making.

Given my current role as Product development manager in Trade finance business in a banks, this project is related to my work, so I will have meaningful use of this project outcome relating to my work as well.

2. Background & Related Work

International Trade plays a critical role in economic development and is closely connected to global energy markets. Oil remains one of the most important commodities influencing global supply chains, transportation networks and industrial production. Previous studies have examined the relationship between oil prices, economic growth and trade balances using traditional econometric

Recent advancements in AI and ML have demonstrated significant improvements in forecasting complex economic and financial time series. However, limited research has explored the use of advanced machine learning models to assess and forecast the import of oil price movements on monthly import and export activity across major global economies.

3. Proposed Solution

The proposed solution is a machine learning-based forecasting framework that predicts monthly import and export values for G20 economies using trade data, oil prices and macroeconomic indicators.
The project will compare multiple machine learning and deep learning models to determine which approach provides the most accurate trade forecasts. The resulting system will generate forecasts, identify key drivers of trade activity and provide insights into how oil price fluctuations influence international trade performance.

Minimum Viable Product (MVP)

- The MVP will include:
- Integrated dataset containing G20 monthly trade data, oil prices and macroeconomic indicators.

- Automated data preprocessing and feature engineering pipeline.
- Forecasting model capable of predicting monthly imports and exports.
- Performance evaluation using standard regression metrics.
- Interactive visualisations showing historical and predicted trade activity
- Preliminary business insights regarding the relationship between oil prices and international trade.

4. Technical Approach

The project will follow a structured machine learning lifecycle consisting of data acquisition, preprocessing, exploratory analysis, feature engineering, model development, evaluation and deployment of results.

Frontend	
Jupyter Notebook visualisations	Interactive charts and dashboards using Matplotlib and Seaborn
Backend	
Python-based data processing and modelling pipeline	Automated feature engineering and forecasting workflows
AI / Machine Learning	
The following models will be implemented and compared: Multiple Linear Regression Random Forest Regression XGBoost	Long Short-Term Memory (LSTM) CNN-LSTM Hybrid Model Transformer-Based Time Series Model (subject to feasibility)
Evaluation Metrics	
R^2 (Coefficient of Determination) MAE (Mean Absolute Error)	RMSE (Root Mean Squared Error) MAPE (Mean Absolute Percentage Error)

5. Data & Resources Required

- UN Comtrade Database – Monthly trade data for G20 economies
- Brent Crude Oil Prices
- West Texas Intermediate (WTI) Oil Prices
- Macroeconomic Indicators
- US Dollar Index (DXY)
- Industrial Production Index
- Inflation indicators (where available)
- Shipping and Freight Indicators
- Baltic Dry Index
- Other publicly available freight indicators

6. Evaluation

The success of the project will be assessed based on forecasting accuracy, model robustness and the ability to generate meaningful business insights.

Success Criteria

- Achieve accurate forecasting of monthly import and export values.
- Demonstrate measurable improvement over baseline models.
- Identify the influence of oil prices and macroeconomic indicators on trade activity.
- Provide meaningful country-level insights across G20 economies.
- Deliver a working forecasting framework suitable for future trade forecasting applications.

7. Project Plan & Timeline

Phase (Weeks)	Key Deliverables
Week 1	Project definition, business problem formulation, literature review and dataset identification
Week 2	Data acquisition from UN Comtrade, FRED and other public sources. Initial data quality assessment
Week 3	Data cleaning, preprocessing, missing value handling and integration of trade, oil and macroeconomic datasets
Week 4	Exploratory Data Analysis (EDA), trend analysis, seasonality analysis, correlation analysis and visualisation development
Week 5	Feature engineering including lag variables, rolling averages, trade balance calculations, growth rates and volatility measures
Week 6	Dataset finalisation, train-validation-test split, baseline model development (Linear Regression and Random Forest)
Week 7	Advanced model development (XGBoost and LSTM implementation), initial model evaluation and performance comparison
Week 8	MVP Completion - Fully functioning forecasting system capable of predicting monthly imports and exports for G20 economies. Initial business insights, visualisations and model evaluation completed
Week 9	Hyperparameter tuning, CNN-LSTM and Transformer model development, model optimisation and commencement of report writing
Week 10	Comparative model evaluation, feature importance analysis, business findings, recommendations, completion of report, documentation and presentation slides
Week 11	Final review, testing, proofreading, formatting, presentation rehearsal and project submission

6.2 Requirements Specification

Requirements Specification

Team Project Module · National College of Ireland · Individual Project

Project Name: Forecasting International Trade Activity using Oil Prices and Macroeconomic Indicators: A machine learning analysis of G20 Economies

Student Name: John Selvin Raja John

Student Number: 25120051

Module / Cohort: HDAIML_SEP25OL

Version: 1

Date: 9 June 2026

Functional Requirements

ID	User Story	Category	Priority	Acceptance Criteria
FR-001	As a student researcher, I want to collect monthly import and export trade data for G20 economies (less Russia), so that a historical forecasting dataset can be created	Data Acquisition	MUST	Given a valid data source, when the data pipeline is executed, then monthly import and export values of all selected countries are downloaded and stored successfully.
FR-002	As a student researcher, I want to import oil price data so that trade activity can be analysed against energy market movements.	Data Acquisition	MUST	Given access to oil price datasets, when the pipeline is executed, then Brent and WTI monthly price data are retrieved and stored successfully.
FR-003	As a student researcher, I want to integrate macroeconomic indicators with trade data so that additional explanatory variables can be included in the forecasting models.	Data Integration	MUST	Given multiple datasets, when pre-processing is completed, then records are merged successfully using a common monthly date field.
FR-004	As a student researcher, I want the system to clean and preprocess raw datasets so that missing values, inconsistencies and formatting issues are resolved.	Data Processing	MUST	Given raw datasets, when preprocessing is completed, then all required variables are standardised and ready for modelling.
FR-005	As a student researcher, I want the system to generate engineered features such as lag variables and rolling averages so that forecasting performance can be improved	Feature Engineering	MUST	Given historical trade and oil price data, when feature engineering is executed, then lag and rolling statistical features are generated successfully.
FR-006	As a student researcher, I want to train baseline machine learning models so that forecasting performance can be benchmarked	Machine Learning	MUST	Given a prepared dataset, when model training is executed, then Linear Regression and Random Forest model are successfully trained and evaluated.
FR-007	As a student researcher, I want to train advanced machine learning and deep learning models so that forecasting accuracy can be improved.	Machine Learning	MUST	Given a prepared dataset when advanced model training is executed, then XGBoost and LSTM models are successfully trained and evaluated.
FR-008	As a student researcher, I want to evaluate forecasting models using standard performance metrics so that model effectiveness can be objectively assessed.	Evaluation	MUST	Given model predictions, when evaluation is performed, then R2, RMSE, MAE and MAPE scores are calculated and displayed.
FR-009	As a student researcher, I want to compare multiple forecasting models so that the most effective forecasting approach can be identified	Evaluation	SHOULD	Given multiple trained models, when evaluation is completed, then model performance results are ranked and compared.
FR-010	As a student researcher, I want to generate visualisations of historical and predicted trade values so that forecasting results can be easily interpreted	Reporting	SHOULD	Given model predictions, when visualisation is executed then charts comparing actual and predicted values are produced.

ID	User Story	Category	Priority	Acceptance Criteria
FR-011	As a student researcher, I want to identify the importance of oil prices and macroeconomic indicators so that the drivers of trade activity can be understood.	Explainability	SHOULD	Given a trained model, when feature importance analysis is performed, then the relative importance of input variables is reported.
FR-012	As a student researcher, I want to forecast future monthly import and export values so that potential trade trends can be analysed	Forecasting	MUST	Given historical data and a trained model, when forecasting is executed, the future monthly trade values are generated successfully.

Non-Functional Requirements

ID	Category	Requirement	Priority	Measurable Target
NFR-001	Performance	The system shall process and prepare the complete dataset efficiently	SHOULD	Data preprocessing completed within 15mins
NFR-002	Performance	Model training shall complete within a reasonable time	SHOULD	Baseline models trained within 5 minutes and advanced models within 45 minutes
NFR-003	Reliability	The data acquisition process shall handle missing data and API interruptions gracefully	MUST	95% successful completion rate for scheduled data downloads.
NFR-004	Accuracy	Forecasting models shall achieve acceptable predictive performance	MUST	Best-performing model achieves R2 =/> 0.80 on the test dataset.
NFR-005	Scalability	The system shall support additional countries and economic indicators without major redesign	SHOULD	Additional datasets can be integrated through the existing pipeline
NFR-006	Usability	Results shall be presented using clear visualisations and summary statistics	SHOULD	All model outputs displayed through charts and evaluation tables.
NFR-007	Maintainability	Code shall follow modular design principles	SHOULD	Data collection, preprocessing and modelling implemented as separate modules.
NFR-008	Reproducibility	The forecasting process shall be repeatable using the same datasets and parameters	MUST	Repeated model runs produce consistent evaluation results within acceptable variance.

Reference

Priority (MoSCoW): MUST = essential for the MVP · SHOULD = important but not vital · COULD = nice to have · WON'T = out of scope this iteration
Functional categories: User Management, Core Features, Data, Integration, UI/UX, Admin
Non-functional categories: Performance, Security, Usability, Compatibility
Acceptance criteria: write as Given [context], When [action], Then [observable result] — each must be checkable as pass or fail

6.3 Interim Report

Interim Report

National College of Ireland · Individual Project

Project Name: Forecasting International Trade Activity using Oil Prices and Macroeconomic Indicators: A machine learning analysis of G20 Economies

Student Name: John Selvin Raja John
Student Number: 25120051
Module / Cohort: HDAIML_SEP25OL

Date: 29 June 2026

1. **Project Overview**

This project develops a machine-learning forecasting tool for monthly import and export trade values across 18 G20 economies. It is intended to show how oil prices, commodity indices and macroeconomic indicators can support trade-activity forecasting beyond simple historical trends.

2. **Deliverables & Progress**

Deliverable	Status	Notes (what remains; will you reach it?)
Project proposal	Done	Submitted, Week 2
Requirements specification and project plan	Done	MUST requirements and MVP scope agreed and documented.
Trade data acquisition and data pipeline (MUST) – FR-001, FR-002, FR-003	Done	UN Comtrade monthly imports/ exports collected for 18 countries; merged with external indicators (variable) data.
Feature engineering and time-based split (MUST) – FR-004	Done	Lagged external variables, country/flow encoding and 2014-2013 training and 2024 test split completed
Implementation of Lag features and rolling-average features as part of Feature Engineering (MUST) – FR-005	Partially complete	Lag variables are implemented. Rolling-average features are not yet in the final pipeline and will be tested in refinement post MVP.
Baseline predictive models (MUST) – FR-008, FR-009, FR-010	Done	Linear Regressor, Random Forest and SVR implemented and evaluated. R^2, RMSE, MAE and MAPE calculated for completed models. Model-comparison and actual-versus-predicted diagnostic charts have been produced.
Identification of the importance of oil prices and macroeconomic indicators (SHOULD) – FR-011	Partially complete	Correlation analysis is available; final feature-importance analysis and interpretation are still to be completed.
MVP Evidence, repository and visualisations	Done	Notebook, model comparison, and plots available ion GitHub repository
Final model refinement (FR-007), evaluation and report	Not started	Will test feature reduction and a tuned HGB and XGBoost benchmark, complete per-country analysis and prepare final documentation

Deliverable	Status	Notes (what remains; will you reach it?)
Forecasting of future monthly import and export values	Not Started	No out-of-sample future monthly forecast has been produced. This requires future external-variable assumptions or scenarios.

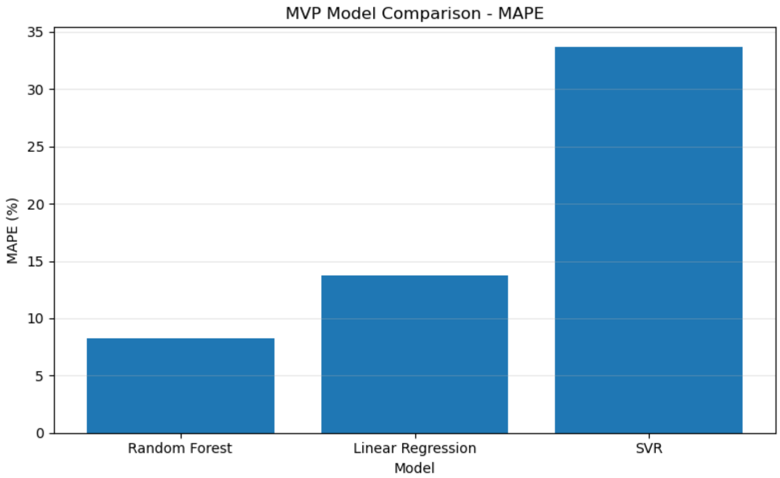
3. Progress Against the MVP

The MVP is substantially complete the core MUST requirements are working. The cleaned modelling dataset contains 4,636 usable monthly country-flow observations after lag creation, with 4,204 observations used for training and 432 observations held out for 2024 testing.

The 2024 test dataset represents approximately USD 28.29 trillion in aggregate import and export trade activity across the 18 selected G20 economies, equivalent to an average of USD 2.36 trillion per month. In the context of this scale, the Random Forest model is currently the strongest baseline model, achieving a mean absolute error (MAE) of USD 5.02 billion and a root mean squared error (RMSE) of USD 8.94 billion. It achieved an R^2 of 0.984 and a MAPE of 8.28%, indicating a strong overall fit and an average percentage forecasting error of approximately 8.3%. Random Forest outperformed both Linear Regression and Support Vector Regression (SVR) across the reported evaluation measures.

4. Evidence / Current State

Data acquired: monthly UN Comtrade import/ export values for 18 G20 economies from January 2014 to December 2024, combined with 12 lagged oil, commodity and macroeconomic indicators. The Figure below show the current model comparison. Random Forest has the lower forecast error.



Repository link: https://github.com/johnslvnjohn/Trade_Forecasting_ML

5. Challenges & Risks

Data coverage differs by country: a small number of monthly records are missing for selected reports. Contingency: retain a documented common time window and report coverage limitations specifically, rather than inputting trade values manually.
The external indicators are highly correlated, which can make feature importance unstable and can inflate a simple model. Contingency: remove redundant features, compare a reduced feature set, and validate results on the fixed 2024 hold-out set.
Model performance is not even across economies, high aggregate accuracy can conceal weaker country-level results. Contingency: include country-level error analysis and avoid claiming that one model is equally reliable for every market.

The final scope will prioritise a defensible forecasting pipeline, evaluation and explanation. Any advanced country-specific modelling that cannot be validated within the final weeks will be treated as future work rather than me including it as an untested feature.

6. Plan to Completion

Week(s)	Planned Work
Wk 8	Complete the remaining MUST requirements by developing and evaluating the advanced forecasting models, including XGBoost, Histogram Gradient Boosting (HGB) and LSTM, and implement future monthly trade-value forecasting
Wk 9	Refine engineered features and compare the baseline Random Forest model with tuned XGBoost, HGB, and LSTM models. Conduct country-level error analysis and feature importance analysis and select the final forecasting approach using the 2024 test set.
Wk 10	Finalise the selected model and complete the results, visualisations, report writing, technical documentation and presentation materials. Test reproducibility, clean and freeze the code, and ensure that the GitHub repository contains all required evidence.
Wk 11	Complete final quality checks, record the video presentation, submit then final project deliverables.

Scope change: The project will retain the 18-country forecasting MVP, but advanced country-specific models will be included only where they produce validated improvements. Otherwise, I will hold them for future work.

6.4 Video Demonstration

Refer below link for the Demo video hosted in YouTube: *https://youtu.be/ZCa0ufGfsS4*